

Matheus Fonseca

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EDUCATION

CUNY Brooklyn College

Bachelor of Science in Computer Science

- **Relevant Coursework:** Data Structures, Discrete Structures, Modern Programming Techniques (OOP), Computer Architecture, Calculus II

Expected June 2029

GPA: 3.93 / 4.0

EXPERIENCE

Hoist AI (NYCSIP × Blackstone Launchpad)

New York, NY

Software Engineer Intern

June 2026 – Aug 2026

- Architected an application-wide automated testing suite, independently defining scope and owning testing decisions end-to-end.
- Implemented Playwright with Pytest to validate frontend UI flows and backend service endpoints across a production platform handling \$3m in freight.
- Proposed evolving the testing suite into an autonomous monitoring agent, outlining methods for severity-based developer alerts and service shutdowns.

Code2Career

New York, NY

Participant

June 2026 – Sep 2026

- Selected as one of 196 BASTA participants from 1,200 applicants for a 10-week mentorship with a Google Software Engineer focusing on data structures, algorithms, and technical interview preparation.
- Solved 50+ coding problems through weekly 1:1 mentorship sessions.

CodePath

Remote

Tech Fellow | AI110 (Applied AI Engineering) and TIP102 (Intermediate Technical Interview Prep)

Feb 2026 – Aug 2026

- Supported instruction for nearly 1,000 students across two courses, driving engagement and comprehension through interactive facilitation covering data structures, algorithms, and AI literacy.
- Mentored students on technical interview preparation, focusing on coding strategies and writing clean, efficient code.

AlgoVerse

Remote

AI Researcher

Aug 2025 – June 2026

- Researched model merging techniques to combine complementary LLMs for improved adversarial robustness.
- Ran model training and evaluations on remote Linux GPU infrastructure.

HONORS / AWARDS

Morgan Stanley CodeToGive Hackathon | 1st Place, Track A

Mar 2026

- Selected as one of 100 from 2,200+ applicants.
- Co-led a team of 8 to build a self-serve volunteer flyering platform for Lemontree, a non-profit providing free food to those in need, enabling volunteers to run campaigns without staff involvement.
- Drove core ideation and developed Citrus, an AI agent that guides volunteers through the flyering process in real time.
- Built leaderboard, get-started page, and onboarding guide; implemented nearby printer functionality and mobile UI optimizations and animations.

COLM Workshop Paper Acceptance: AIW & AdvML-Frontier

Jul 2026

- Co-authored a paper accepted for presentation at two COLM 2026 workshops, contributing to experimental design, implementation, result analysis, and paper writing.
- Researched adversarial fine-tuning datasets, identifying and integrating the primary datasets into the training pipeline.
- Fine-tuned Gemma-3-1B-it adversarial jailbreak models, including a 9K-sample experimental variant, and conducted model training and evaluation.
- Conducted weight-space analysis and model merging experiments to evaluate the impact of safety fine-tuning on model behavior and performance.

PROJECTS

SendToMac | Python, Supabase, rumps, Swift

- Developed a **cross-platform application** enabling asynchronous transfer of text and URLs from iOS to macOS, even while the Mac is offline.
- Implemented **background macOS client** using **Python**, **AppKit**, and **rumps** to display real-time alerts with user actions (Open, Snooze, Ignore).
- Prototyped iOS interface using **SwiftUI** with AI assistance to accelerate design and implementation.
- Enabled content transfer via **REST API** with responsive success/failure alerts.
- Integrated scheduling and expiration logic using timestamps to manage message flow and cleanup.

Chess Web-App | HTML, CSS, JavaScript

- Developed an interactive **chess web application** with full rules enforcement, including check, checkmate, stalemate, and legal move validation.
- Implemented real-time piece selection with visual highlights and move restrictions based on king safety and game rules.
- Built modular move logic and game state tracking to ensure accurate gameplay flow and turn handling.
- Designed a responsive, user-friendly interface using **HTML**, **CSS**, and **JavaScript**.

SKILLS

Languages: Python, Java, C; Familiar with: HTML/CSS, JavaScript

Frameworks/Tools: Flask, SQLite, Supabase, Git, REST APIs, Pytest, Playwright, PyTorch, Hugging Face, Transformers, Linux

Spoken Languages: English (Native), Portuguese (Native), Spanish (Basic)